2.1 理论计算题

1.解：$a \cdot b = 2\times1+(-1)\times4+3\times(-2) = 2-4-6 = -8$

2.解：$C = A\times B = \begin{bmatrix} 1\cdot2+0\cdot0+2\cdot3 & 1\cdot1+0\cdot(-1)+2\cdot2 \\ (-1)\cdot2+3\cdot0+1\cdot3 & (-1)\cdot1+3\cdot(-1)+1\cdot2 \end{bmatrix}$

$C = \begin{bmatrix} 8 & 5 \\ 1 & -2 \end{bmatrix}$

3.解：$\|a\|_F = \sqrt{2^2+(-1)^2+3^2} = \sqrt{14}$

3.1理论计算题

解：定义事件：

$A$：患病，$P(A)=0.001$

$B$：检测阳性

灵敏度：$P(B|A)=0.99$

假阳性率：$P(B|\bar{A})=0.02$

由贝叶斯公式得：

$$P(A|B) = \frac{P(B|A)P(A)}{P(A)P(B|A)+P(B|\bar{A})P(\bar{A})} = \frac{0.99\times0.001}{0.99\times0.001+0.02\times0.999} = \frac{0.00099}{0.02097} \approx 0.0472$$

综上所述，一个人检测结果为阳性，他真正患病的概率约为4.72%

4.1理论计算题

解：$\because z = (w_1x_1+w_2x_2-y)^2$，其中 $x_1=2, x_2=1, y=3$

(1) 令 $u = w_1x_1+w_2x_2-y$，则 $z=u^2$

$$\frac{\partial z}{\partial w_1} = 2u\frac{\partial u}{\partial w_1} = 2(w_1x_1+w_2x_2-y)\cdot x_1$$

$$\frac{\partial z}{\partial w_2} = 2(w_1x_1+w_2x_2-y)\cdot x_2$$

(2) 当 $w_1=0.5, w_2=1$ 时，

$u = 0.5\times2+1\times1-3 = -1$

则：$\frac{\partial z}{\partial w_1} = -4, \frac{\partial z}{\partial w_2} = -2$

5.1理论计算题

解：$\because$ 线性回归模型：$y=wx+b$

均方误差损失：$L = \frac{1}{n}\sum_{i=1}^{n}(y_i-(wx_i+b))^2$

对 $w$ 求偏导：

$$\frac{\partial L}{\partial w} = \frac{1}{n}\sum_{i=1}^{n}2(y_i-wx_i-b)\cdot(-x_i) = -\frac{2}{n}\sum_{i=1}^{n}x_i(y_i-wx_i-b)$$

对 $b$ 求偏导：

$$\frac{\partial L}{\partial b} = \frac{1}{n}\sum_{i=1}^{n}2(y_i-wx_i-b)\cdot(-1) = -\frac{2}{n}\sum_{i=1}^{n}(y_i-wx_i-b)$$

6.1 理论计算题

1. 解：∵ 样本 $x_1, x_2, \cdots, x_n$ 独立同分布于正态分布 $N(\mu, \sigma^2)$

∴ 似然函数 $L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$

2. 证明：

对似然函数取对数：

$$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

对 $\mu$ 求偏导并令其为0：

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$\therefore \sum_{i=1}^{n} x_i - n\mu = 0$$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

3. 证明：

令 $v = \sigma^2$，对 $\sigma^2$ 求偏导：

$$\left(\frac{\partial L(\mu, v)}{\partial v} = \right)_x$$

$$\frac{\partial \ln L(\mu, v)}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

代入 $\hat{\mu}$ 并解：

$$-\frac{n}{2\hat{\sigma}^2} + \frac{1}{2(\hat{\sigma}^2)^2}\sum_{i=1}^{n}(x_i-\hat{\mu})^2 = 0$$

两边乘以 $2(\hat{\sigma}^2)^2$ 得：

$$-n\hat{\sigma}^2 + \sum_{i=1}^{n}(x_i-\hat{\mu})^2 = 0$$

$$\therefore \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2$$